

Laboratory Session #2

Deliverables

This assignment is to be completed in pairs and will be submitted to Crowdmark (PDF). Pairing allocations are posted in the lab session and can also be found on LEARN.

In this assignment, you will:

- Decipher code written by others
- Develop helpful input and output statements
- Insert concise, descriptive comments
- Create arrays, use basic array functions, and perform mathematical operations with arrays

Your Crowdmark submission should contain:

1. A header comment block, formatted as follows:
% Authors: Names of those working on the assignments
% Purpose: Short description of what the code does
% Date Created: Today's date
% Acknowledgements: Any outside sources used to complete
% assignment, including website, books, and people (other
% than the teaching team)
2. The output of the original code for each test case
3. The code that you wrote and its output for each test case
4. Your answers to the questions included in the assignment.

Crowdmark Submission with Partners

Prior to submitting your assignment work to Crowdmark, you need to tell Crowdmark that you are part of a group. To do this, complete the following steps:

- Navigate to the Lab Assignment submission area in Crowdmark.
- Click the "Add Group Members" button at Crowdmark and choose your group for this assignment. You can also wait for someone else to add you to their group. Note that after the assignment has been submitted, your group cannot be changed.
- Each group member you add will receive an email notification and shared access to this page.
- Any group member may submit the assignment or edit group members.

Lab Exercise

The script below is a fully functional program that plays a dice game. Unfortunately, the person who wrote the code left it in a state where it is very difficult to interpret the rules of the game. Run the program and work line by line to figure out the rules that the game operates by. Then rewrite the code using a more helpful style, to communicate to both the users and the developers what this code is for.

```

1  %3 round dice game
2  xx = [1 2];
3
4  s = input('Start: ');
5  rng(s)
6  p1 = [randi([1,6]), randi([1,6]), randi([1,6])]; p12 = [randi([1,6]), ...
7      randi([1,6]), randi([1,6])];
8      p2r1 = [randi([1,6]), randi([1,6]), randi([1,6])];
9      p2r2 = [randi([1,6]), randi([1,6]), randi([1,6])];
10 g = [sum(rem(abs(p1 - p12),2)),sum(rem(abs(p2r1 - p2r2),2))];
11 gb = g + [sum(p1+p12)>2*sum(p2r1+p2r2) sum(p2r1+p2r2)>2*sum(p1+p12)];
12 f = gb * (gb(1) ~= gb(2));
13 [val, ID] = max(f);
14 fprintf("Result = %d. If 0, tie; else P%d.\n", val, xx(ID))

```

Hints

- Using the same seed as input will allow you to produce the same results each time.
- To get more information about what a function does, type 'help XXX' on the command line with respect to a specific function, or search for the function on Mathworks' documentation website.
- Check to make sure that your revised code gets the same results as the original code.

Questions to Consider

- How can the code be broken into clearer sections?
- Should any intermediate steps in the game be reported to the user? Which ones?
- How should the rules of the game be communicated to the user?
- How should the logic of the code be communicated to the developer?
- Are the inputs and outputs of the code sufficiently descriptive?

Test Cases

1. Input → 4567
2. Input → 9998
3. Input → 19

What you need to submit

To Crowdmark:

- The code for your refactored version and the resulting command window from each test case.
- A short (2-3 sentence) description of the rules of the game.
- A short list (5-10 bullets) of the design decisions that went into reformatting the code, particularly in response to the questions outlined above.